MERT TURKEL

Computer Engineering Student & Software Developer (AI, Backend & Cybersecurity)

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PROFESSIONAL SUMMARY

Results-driven Computer Engineering & Management Information Systems student and Software Developer specializing in scalable backend architectures, AI-integrated SaaS platforms, cross-platform mobile applications, and application security. Proven track record of winning hackathons (3rd Place at Cursor Hackathon), leading TUBITAK BiGG & Teknopark Istanbul pre-incubation ventures, and holding certified expertise in Application and API Security (CAP V2 with Merit, APIsec Certified Practitioner). Adept at leveraging Go, Python, Rust, C/C++, Flutter, and Next.js to engineer high-concurrency, privacy-first, and security-hardened software systems while spearheading university-wide engineering communities.

EDUCATION

Istanbul Health and Technology University (ISTUN) — B.Sc. in Computer Engineering | *Sep 2025 – Present (Sophomore / 2nd Year)* | *Istanbul, Turkiye*

- **Relevant Focus:** Data Structures & Algorithms, Object-Oriented Programming (C/C++), Software Architecture, Systems Programming, Cybersecurity.

Istanbul University (AUZEF) — B.Sc. in Management Information Systems (Distance Education) | *Jul 2026 – Present* | *Istanbul, Turkiye*

American Street English — Professional English Program (B1/B2 Upper-Intermediate) | *Oct 2025 – Present* | *Proficient in Technical Documentation Translation*

Elazig Private Simya College — High School Diploma (Science & Mathematics Track) | *Graduated May 2025* | *Elazig, Turkiye*

TECHNICAL PROJECTS

LOOP — AI-Powered B2B SaaS Logistics & Delivery Platform (TUBITAK BiGG Candidate) | *2025 – Present* | *Go, Python, Flutter, Docker, REST APIs, Voice AI* | *github.com/mertturkel234/loop_app*

- **Architected** an end-to-end B2B SaaS delivery management platform from scratch using Go, Python, and Flutter to unify field operations into a centralized real-time command hub, advancing to the TUBITAK BiGG & Teknopark Istanbul company formation stage.
- **Engineered** an autonomous courier-assignment and route-optimization algorithm that dynamically matches orders with field couriers, significantly reducing manual dispatch overhead and delivery latency.
- **Integrated** a hands-free, voice-enabled AI Logistics Assistant within the Flutter mobile client to enable real-time operational status updates and safe on-road navigation for couriers.

UrbanPulse — Privacy-First Smart City Intelligence Platform (3rd Place — Cursor Hackathon) | *Jun 2026* | *Go, WebSocket, REST API, Docker, Render.com* | *city-intelligence-dashboard.vercel.app*

- **Secured 3rd Place** at the Cursor Hackathon by designing and deploying a high-throughput, privacy-first municipal data platform capable of real-time urban traffic and crowd analytics.
- **Developed** a GDPR- and KVKK-compliant backend pipeline in Go that irreversibly anonymizes personally identifiable information (faces and license plates) prior to executing object detection on live camera streams.
- **Implemented** low-latency WebSocket data streams and containerized RESTful microservices with Docker on Render.com to deliver sub-second telemetry updates to the municipal intelligence dashboard.

CoreSync AI — Autonomous B2B Technical Pre-Sales & API Integration Platform | *Oct 2026* | *Python, FastAPI, Next.js, TypeScript, PostgreSQL (pgvector), Redis, Docker* | *github.com/mertturkel234/coresync-ai*

- **Built** an AI-powered B2B API onboarding platform using FastAPI, Next.js (App Router), and PostgreSQL (pgvector) that ingests vendor API documentation and generates grounded Python integration code.
- **Designed** a sandboxed execution engine separating LLM-generated code suggestions from verified runtime execution results against controlled demo APIs, accelerating developer onboarding and pre-sales validation.

LaundroQR — IoT-Based Multi-Tenant Smart Dormitory Laundry Ecosystem | *Oct 2026* | *Flutter, Next.js 15, Supabase (PostgreSQL + RLS), IoT/QR* | *github.com/mertturkel234/LAUNDROQR*

- **Developed** a full-stack smart laundry management ecosystem featuring a cross-platform Flutter resident application and a Next.js 15 admin dashboard for university dormitories and co-living spaces.
- **Enforced** strict multi-tenant data isolation using Supabase PostgreSQL Row-Level Security (RLS) and eliminated double-booking race conditions via atomic session locking (FOR UPDATE) and partial unique indexing.

SentraEDR — Lightweight Rust-Based Endpoint Detection & Response Platform | *2026* | *Rust, Tokio, Windows ETW, MITRE ATT&CK, Behavioral Heuristics*

- **Engineered** a modular, low-memory Endpoint Detection and Response (EDR) system in Rust utilizing asynchronous Tokio runtimes and Event Tracing for Windows (ETW) for real-time kernel and process telemetry collection.
- **Implemented** multi-signal heuristic risk scoring aligned with the MITRE ATT&CK framework to detect Remote Access Trojans (RATs), info-stealers, and malicious PowerShell executions with automated safe remediation.

Sign2Voice — Offline AI Sign Language Translation Mobile Platform | *Oct 2026* | *Flutter, Dart, TensorFlow Lite (TFLite), MediaPipe 3D*

- **Created** an accessibility-focused mobile application in Flutter capable of translating sign language gestures into synthesized speech completely offline with zero network dependency.
- **Optimized** on-device MediaPipe 3D hand-tracking pipelines and quantized TFLite inference models to achieve real-time, low-latency gesture recognition on mobile hardware.

AeroNode & VitaxRing — Autonomous Drone Mesh Network & Smart Wearable R&D Ventures | *Oct 2026* | *C/C++, STM32, Raspberry Pi CM4, MQTT, Embedded Systems*

- **Designed** AeroNode, a decentralized autonomous drone mesh-networking software platform utilizing MQTT and embedded Linux (Raspberry Pi CM4) to sustain resilient disaster-relief logistics and emergency communications.
- **Spearheaded** VitaxRing as Solo Founder, architecting an indigenous smart-ring wearable prototype using STM32 microcontrollers targeted for the Teknopark Istanbul Pre-Incubation Program.

Process Automation Suite — Autonomous Reporting Assistant & English Study Automation | *2025 – 2026* | *Python, Telegram Bot API, python-docx, Audio Synthesis* | *github.com/mertturkel234/Weekly-reporting-Telegram-bot*

- **Automated** weekly engineering status reporting by building a command-driven Python Telegram Bot that parses team updates and instantly compiles standardized Microsoft Word (.docx) documents, reducing manual reporting time by 80%+.
- **Developed** an automated Python pipeline that processes the Oxford 3000 vocabulary corpus into structured audio podcasts to accelerate language acquisition.

LEADERSHIP & EXPERIENCE

IEEE ISTUN Student Branch — President of Coordination & Student Clubs (KOK) | *Oct 2026 – Present* | *Co-Founder & Vice Chair, IEEE ISTUN CyberSec Committee (Sep 2025 – Oct 2026)*

- **Promoted** to KOK President to oversee cross-committee technical operations, strategic partnerships, and student engineering initiatives across the university.
- **Co-founded** the university's official Cybersecurity Committee, designing and directing the university's highest-attendance Capture The Flag (CTF) Workshop to train students in practical offensive/defensive security.

Cybersecurity Industry Panel ('Current Threats & Defense Methodologies') — Project Leader & Operations Manager | *Apr 2026* | *Istanbul, Turkiye*

- **Orchestrated** an end-to-end university cybersecurity summit from zero to execution, hosting senior Security Architects from Coca-Cola Icecek (CCI) and managing all speaker relations, stage operations, and campus outreach.

ISTUN TEKNOFEST Club — Vice President & Editorial/Administrative Board Member | *Jan 2026 – Present* | *Istanbul, Turkiye*

- **Direct** administrative operations and competition team coordination while authoring and editing the monthly National Technology Bulletin on autonomous systems, defense technologies, and AI innovation.

T3 Foundation (Turkiye Teknoloji Takimi Vakfi) — Warehouse & Logistics Operations Volunteer | *Oct 2026 – Present* | *Istanbul, Turkiye*

- **Executed** end-to-end warehouse inventory management, equipment dispatch, and field supply-chain logistics operations to ensure seamless operational readiness for national technology festivals (TEKNOFEST) and educational programs.

Ecosystem & Community Leadership Roles | *2026 – Present* | *Istanbul & Elazig, Turkiye*

- **Brand Ambassador — Teknopark Istanbul (Mar 2026 – Present):** Represent Turkiye's premier deep-tech R&D hub on campus, connecting student founders with incubation programs, career days, and R&D center tours.
- **Project Development Committee Member — Turkiye Girisim Agi (Jun 2026 – Present):** Evaluate early-stage startup concepts and co-develop scalable project roadmaps across national university chapters.
- **University Ambassador — Genc TUYAFED (Mar 2026 – Present):** Coordinate youth federation initiatives and inter-university technology networking events.
- **Customer Consultant & Operations Coordinator — Karabalik Shopping Center (Prior to Sep 2025):** Managed high-volume floor operations and customer conflict resolution, honing rapid decision-making and crisis management under pressure.

TECHNICAL SKILLS

- **Programming Languages:** Go (Golang), Python, C, C++, Rust, Dart, Java, JavaScript, TypeScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** Flutter, FastAPI, Next.js, React, Tailwind CSS, Framer Motion, TensorFlow Lite (TFLite), MediaPipe
- **Databases, Cloud & Infrastructure:** PostgreSQL (pgvector, RLS), Supabase, Redis, Docker, RESTful APIs, WebSocket, Linux (Kali, Ubuntu/WSL), GitHub Actions
- **Cybersecurity & AppSec:** Application Security (CAP), API Security, Endpoint Detection & Response (EDR), MITRE ATT&CK, Vulnerability Analysis, Wireshark, Burp Suite, Metasploit, GDPR/KVKK Anonymization
- **Hardware, Embedded & Tools:** STM32 Microcontrollers, Raspberry Pi CM4, MQTT Protocol, Git/GitHub, VS Code (GitHub Copilot), Cursor, Android Studio, IntelliJ IDEA

CERTIFICATIONS, HONORS & CONFERENCES

- **Certified AppSec Practitioner V2 (CAP) — Achieved with 'Merit'** | The SecOps Group (May 2026)
- **APIsec Certified Practitioner (ACP) — 92% Exam Score** | APIsec University (Jan 2026)
- **3rd Place Winner — Cursor Hackathon** | UrbanPulse Privacy-First Smart City Platform (Jun 2026)
- **Project Exhibitor & Developer — IEEE ISTUN CS Fair** | IEEE ISTUN Student Branch (May 2026)
- **Delegate — CSCON'26 (IEEE Computer Society Congress)** | Isik University — Quantum Computing (Qiskit), Go Backend & Cybersecurity (Mar 2026)
- **Participant — Teknopark Istanbul Career Days & R&D Center Tours** | Teknopark Istanbul (May – Oct 2026)